Madan Y N

Bengaluru, India | madanyn3@gmail.com | 7795 270 926 | madanyn3.github.io | madanyn3 | madanyn3

Education

IIT Bombay *Nov 2020 - May 2024*
B.Tech Honours in Electrical Engineering & a minor in Computer Science and Engineering

- GPA: 8.71/10
- **Coursework:** Advanced Computer Architecture, Algorithmic Design of Digital Systems, Design & Analysis of Algorithms, Linear Algebra, Partial Differential Equations, Computer Networks

Work Experience

Silicon Design Engineer II *Bengaluru, India*
AMD | Server CPU Performance Modeling Group *July 2024 – Present*

- Model and analyze new cache-hierarchy features in AMD's internal **cycle-accurate simulator**, performing bottleneck analysis and design space exploration
- Modeled the switch connecting cores to the data fabric, reducing absolute bandwidth error against the RTL model from ∼15% to **1.64%** under directed stress testing
- Own and maintain the infrastructure for **heterogeneous multi-core simulation** on a compute cluster, enabling studies that mix different core architectures; used by 3+ performance teams across AMD
- Explored **reinforcement-learning-based scheduling** (Q-learning) to improve L2 cache bandwidth utilization, achieving a ∼1% geomean bandwidth gain over a production-tuned baseline scheduler across **SPEC CPU** workloads; findings informed next-generation scheduler design
- Implemented a detailed **asynchronous interface model** enabling simulation of two modules at different clock frequencies; identified new downstream bottlenecks and recommended queue-size tunings

Research Experience

Analysis of Contemporary ISA Architectures | IIT Bombay *June 2024*
Guide: Sunil Shenoy, VP Emeritus Intel & Prof. Laxmeesha Somappa, IIT Bombay

- Quantified the role of **ISA** in microarchitecture performance on SPEC 2017 workloads using **Gem5**
- Modeled the Intel Golden Cove P-Core microarchitecture from reverse-engineering documents
- Identified Gem5 modeling limitations and explored approximations to bridge the gap

Modeling Performance of Transformers on FPGA | TCS Research *Jul 2023*
***Best Poster Award**, IEEE HiPC Student Research Symposium (Dec 2023)*

- Developed two **low-latency** pipelined architectures and formulated latency expressions for GPT-2 inference
- Conducted design space exploration using a **genetic algorithm** to determine efficient resource partitioning
- Estimated **21× and 119× speedups** over a dual-core i5 CPU baseline using performance models constrained by Alveo U280 resources

Clock-Gating in Low-Power Microcontrollers | RnD Project IITB *May 2024*
Guide: Prof. Madhav Desai, IIT Bombay

- Evaluated the power reduction from **clock-gating** a multi-cycle non-pipelined RISC-V microcontroller
- Emulated a RISC-V RV32I μArch in Python, generated VHDL with the Ahir **LLVM-to-HDL** toolchain, verified functional correctness, enabled clock-gating and synthesized the design

Projects

Cardiac Redundancy: Auxiliary Pump Design Study | Self Project *June 2026 – Present*

- Investigating whether a hypothetical auxiliary heart could sustain critical-organ oxygen delivery after native cardiac arrest, using mathematical modeling and simulation to evaluate circulatory support strategies

- Formulated a constrained design study under finite oxygen and fuel budgets; characterized auxiliary-pump behavior across **1,500 static parameter combinations** and **10 dynamic scenarios**
- Extended the published Smith et al. **0D two-ventricle model** in Python with organ-specific vascular conductances, oxygen transport and glucose–glycogen accounting; implemented stiff ODE integration, nonlinear root finding and event-driven valve dynamics
- Verified numerical behavior through conservation checks, solver comparisons, tolerance refinement and a **191-test suite**; identified coronary oxygen deficits in **all 6 joint hydraulic–metabolic screening scenarios**. Independently designed the study, audited AI-assisted implementation and authored the analysis

Analytical Edge-ML Performance Simulator | **Self Project** *Sept 2026 – Present*

- Developing a C++20 performance simulator for quantized neural-network kernels, deriving operation counts and ordered memory accesses from a pinned CMSIS-NN int8 fully connected implementation.
- Implemented configurable set-associative LRU cache hierarchies, pipelined memory-service models, and analytical acceleration for regular access patterns; checked counts, completion times, and final states against independent reference models.

Tile-Aware Kernel Optimization | **Self Project** *Jan 2026*

- Developed experimental OpenCL kernels for 16×16 tiled GEMM and row-tiled streaming softmax, exploring local-memory reuse and parallel reductions for transformer attention.
- Built an attention benchmark harness covering 48 problem shapes, recording kernel execution time, host-observed pipeline time, and numerical error against PyTorch references.
- Prototyped C++ AVX-512 FP32 matrix-multiplication microkernels using vector broadcasts and fused multiply-add intrinsics.

Technical Skills

Languages: C++, Python, C, Bash, VHDL, Verilog, x86 Assembly
Parallel & ML Frameworks: OpenCL, PyTorch, TensorFlow, ONNX Runtime
Simulation & Tools: Gem5, cycle-accurate performance simulators, Git, Linux (bare-metal and WSL)
Scientific Python: NumPy, Pandas, Matplotlib, OpenCV

Selected Writing

Why Apple Silicon is Better (Jan 2025): Analysis of Apple M1 and Intel Golden Cove cache hierarchy, instruction fetch, decoding and memory ordering.

Life in Disequilibrium (Nov 2025): Science essay on cellular bioenergetics and metabolism, drawing on Nick Lane's *The Vital Question* and *Transformer*.

Writing portfolio: madanyn3.github.io/blogs/

Awards & Honours

Best Poster Award, IEEE HiPC Student Research Symposium *(Dec 2023)*
JEE Main 2020: 99.64 percentile (0.92 million participants); **JEE Advanced 2020:** 96.48 percentile (0.16 million) *(2020)*
Qualified for the **Kishore Vaigyanik Protsahan Yojana** (KVPY-SA and KVPY-SX) fellowship *(2018 & 2019)*

Position of Responsibility

Team Leader | **ExoFly IIT Bombay** *May 2023 – April 2024*
Student team building a manned VTOL aerial vehicle with a 90 kg payload capacity

- Led a **30+ member** team building a safe and compact VTOL manned aerial vehicle
- Secured and managed annual funding of **INR 1.4 million** for the first iteration of the vehicle
- Represented IIT Bombay in the **UAS Challenge** (Institution of Mechanical Engineers) and won the fastest-time-to-build drone category